Analytical Synopsis: Customer Base Audit

1. Executive Summary

This project provides a comprehensive audit of a company's customer base using transaction data. By segmenting customers into deciles and tracking historical cohorts, the analysis identifies key drivers of revenue and evaluates customer retention, providing actionable business intelligence.

2. Decile Analysis and Revenue Concentration

The customer base was divided into ten equal segments (deciles) based on profitability. The analysis revealed a significant concentration of value: the top 10% of customers (Decile 1) accounted for approximately 39.6% of total revenue and 39.8% of total profit. These premium customers exhibit an Average Order Frequency (AOF) of 5.5 and an Average Order Value (AOV) of 7,367, far exceeding the bottom deciles. This suggests that marketing efforts should be disproportionately focused on retaining this vital segment.

3. Cohort Analysis and Retention

To understand customer lifecycle and retention, we conducted a cohort analysis grouping customers by their acquisition quarter.

* Pre-2021 Cohort: Remains a strong foundation, generating consistent revenue (e.g., 55.4M in 2024), though showing expected gradual attrition.

* Recent Cohorts (2021-2022): The analysis tracks the percentage of active customers quarter-over-quarter. For instance, the Q1 2021 cohort started with high activity but demonstrated a sharp drop-off in subsequent quarters, stabilizing at around a 10-16% active rate. This indicates a potential need for targeted re-engagement campaigns immediately following a customer's first quarter.

4. Data Visualization

To effectively communicate the distribution of key variables, data was visualized using Histograms, Polygons, and Ogives (Cumulative Frequency). These visual tools are essential for quickly identifying the central tendency and spread of customer behaviors, such as order values or waiting times. Additionally, scatterplots were utilized to identify categorical trends and avoid aggregate misinterpretations like Simpson's Paradox.

5. Conclusion

The audit demonstrates that the company relies heavily on a small segment of high-value customers. Future strategies should prioritize maximizing the AOV and retention rates of mid-tier deciles to reduce reliance on the top 10%, while implementing early-lifecycle interventions to improve the long-term retention of new cohorts.